

YUNZHE GAO

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EDUCATION

Nanyang Technological University (NTU)

Singapore

M.Sc. in Computer Control and Automation

Aug. 2025 – Jan. 2027 (Expected)

- **GPA: 4.75 / 5.0**
- Honors: EEE Global Merit Award.
- Advisor: Prof. Mao Kezhi
- Dissertation Topic: Design and Deployment of Lightweight Large Language Models (LLM) for Efficient, Localized Deployment.
- Selected Coursework: Robotics & Intelligent Sensors, Machine Vision, Video Signal Processing.

Northeastern University (NEU)

Shenyang, China

B.E. in Automation

Sept. 2021 – Jul. 2025

- **GPA: 3.76 / 5.0 (Avg. Score: 87.6/100)**
- Honors: University Academic Scholarship.
- Selected Coursework: Introduction to Intelligent Control (97/100), Motion Control System (91/100), Artificial Intelligence Foundation (93/100).

RESEARCH EXPERIENCE

Energy-Efficient Edge-Cloud Collaborative LLM Inference

NTU, Singapore

Master Dissertation (Advisor: Prof. Mao Kezhi)

Aug. 2025 – Present

- Designed "Green-Edge-Cloud," an adaptive collaborative inference framework that dynamically routes queries between a lightweight edge Student model (Qwen-1.5B) and a cloud Teacher model (Qwen-72B) via training-free, zero-shot confidence scores.
- Conducted Supervised Fine-Tuning (SFT) via LoRA on the 1.5B Student model using 6,700 high-quality Chain-of-Thought (CoT) trajectories distilled from the 72B Teacher, significantly improving logical reasoning and calibrating confidence distributions.
- Implemented real-world energy measurement using CodeCarbon to track GPU/CPU power consumption, rigorously modeling cloud-offloading network latency and hardware idle-state energy.
- Achieved a superior Pareto frontier on GSM8K and SVAMP benchmarks; recovered over 50% of the accuracy gap while reducing total energy consumption by 81.16% and accelerating inference speed by 5.6x compared to cloud-only baselines.

Democratizing Fine-Grained Manipulation via Imitation Learning

NEU, China

Undergraduate Thesis (Advisor: Prof. Li Zhi)

Nov. 2024 – Jun. 2025

- Developed a low-cost, 6-DOF mobile teleoperation system (Koch v1.1, under \$250) integrating the Action Chunking Transformer (ACT) and Conditional VAE (CVAE) to mitigate compounding errors in continuous control.
- Replaced the standard ResNet backbone with MobileNetV2 and integrated YOLOv5 for dynamic object segmentation, reducing inference latency to 0.014s (20% faster) and enabling real-time 60Hz control.
- Successfully deployed the framework in real-world scenarios, achieving an 86% grasping success rate in high-precision plug insertion tasks and a 96.3% success rate in dynamic logistics sorting tasks.

Coal Ash Composition Prediction via Machine Learning

NEU, China

Project Lead (First Author)

Jan. 2024 – Aug. 2024

- Proposed a novel inversion method using Particle Swarm Optimization - Extreme Learning Machine (PSO-ELM) to predict coal ash content from Near-Infrared (NIR) spectral data.
- Designed a hybrid feature selection pipeline combining Savitzky-Golay (SG) smoothing and Random Frog (RF) algorithms, effectively reducing spectral redundancy.

- Achieved state-of-the-art prediction accuracy with an R^2 of 0.98 and RMSE of 0.79, validated on real-world industrial datasets.
- Outcomes: Published findings as the First Author in Control Engineering of China (Core Journal).

Digital Twin-based Mineral Processing Intelligent Decision System

NEU, China

Core Algorithm Engineer

Jan. 2023 – Nov. 2023

- Engineered the “Abnormal Pattern Retrieval System (ECAS)” using Digital Twin technology to monitor massive industrial equipment data.
- Implemented correlation analysis algorithms to retrieve abnormal operational patterns, significantly enhancing fault detection efficiency.
- Outcomes: Awarded the Grand Prize at the NEU Innovation and Entrepreneurship Competition, the National Silver Award at the "Internet+" Competition (Top 1% nationwide), and secured 2 Software Copyrights.

PUBLICATIONS

- [1] **Y. Gao**, J. Zou, D. Xiao. “Prediction of Coal Ash Composition Based on Near-Infrared Visible Spectroscopy and PSO-ELM.” *Control Engineering of China*, 2024. (**Core Journal**)
- [2] **Y. Gao**, K. Mao. “Green-Edge-Cloud: Energy-Efficient Edge-Cloud Collaborative LLM Inference.” *Manuscript in preparation* (Targeting **ICML 2026 Workshop**).

INTELLECTUAL PROPERTY

- [1] **Mineral Processing Equipment Abnormal Pattern Retrieval and Index Correlation Analysis System (ECAS) V1.0**
K. Zhang, **Y. Gao** (2nd Author). *Computer Software Copyright (China)*, Reg. No. 2023SR1469715, Nov. 2023
- [2] **Intelligent Venue Resource Allocation and Security Operation Visual Analysis Software (IVVAS) V1.0**
K. Zhang, P. Zhu, **Y. Gao** (3rd Author). *Computer Software Copyright (China)*, Reg. No. 2023SR1421636, Nov. 2023

SELECTED AWARDS

- **Global Merit Award**, Nanyang Technological University 2025-2026
- **National Silver Award (Top 1% nationwide)** Dec. 2023
China International College Students' Innovation Competition
- **University Academic Scholarship**, Northeastern University 2024-2025
- **University Academic Scholarship**, Northeastern University 2021-2022

TECHNICAL SKILLS

- **Programming & Languages:** Python, C/C++, MATLAB, SQL, LaTeX.
- **AI & LLM Frameworks:** PyTorch, Hugging Face (Transformers, PEFT/LoRA, TRL), vLLM, CUDA, CodeCarbon, Scikit-learn.
- **Robotics & Simulation:** ROS (Robot Operating System), MuJoCo, Gazebo, MobileNet/YOLO.
- **Systems & Deployment:** Linux, Docker, Git, Edge Computing Optimization.
- **Core Algorithms:** LLM Quantization & Knowledge Distillation, Action Chunking Transformer (ACT), Reinforcement Learning, PSO-ELM (Swarm Intelligence), Digital Twin.

PROFESSIONAL ASSOCIATIONS

- **Student Member**, China Computer Federation (CCF) 2024